Section 3.9

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1) $L(x) = f(2) + f'(2)(x-2)$
$= (2^3 - 2(2)+3) + (3(2^2)-2)(x-2)$
$= 7 + 10x - 20$
$= 10x - 13$

6. a) $L(x) = \sin 0 + \cos 0\,(x-0)$
$= 0 + 1(x) = x$

b) $L(x) = \cos 0 + (-\sin 0)(x-0)$
$y = 1 + 0 \cdot x = 1$

c) $L(x) = \tan 0 + (\sec^2 0)(x-0)$
$= 0 + 1 \cdot x = x$

17. $y = x^3 - 3\sqrt{x}$
$dy = \left(3x^2 - \frac{3}{2\sqrt{x}}\right)dx$

18. $y = x\sqrt{1-x^2}$
$dy = \left(\sqrt{1-x^2} + x\left(\frac{1}{2\sqrt{1-x^2}} \cdot (-2x)\right)\right)dx$
$= \left(\sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}}\right)dx$

29. a) $\Delta f = f(1.1) - f(1) = (1.1^2 + 2(1.1)) - (1^2 + 2(1)) = 3.41 - 3 = 0.41$
b) $df|_{x=1} = f'(1)dx = (2(1)+2)(0.1) = 0.4$
c) $|\Delta f - df| = |0.41 - 0.4| = 0.01$

37. $dS \approx 6(x_0+dx)^2 - 6x_0^2$
$= 6x_0^2 + 12x_0 dx + 6(dx)^2 - 6x_0^2$
$= 12x_0 dx + \cancel{6(dx)^2} = 12x_0 dx$

42. $dr = \Delta r = \frac{5}{2\pi}$
$dD = d(2r) = 2dr = \frac{5}{\pi}$ (Increase in diameter)
$dA = d(\pi r^2) = 2\pi r\,dr = 2\pi r \cdot \frac{5}{2\pi} = 2\pi \cdot \frac{25}{2} \cdot \frac{5}{2\pi} = \frac{125}{2}$ (Increase in cross-sectional area)

51. From $W = a + \frac{b}{g}$ ; $a, b$ const
$dW = -\frac{b}{g^2}\,dg$
$\frac{dW_{moon}}{dW_{earth}} = \frac{-b/g_{moon}^2\,dg}{-b/g_{earth}^2\,dg} = \frac{g_{earth}^2}{g_{moon}^2} = \frac{9.8^2}{1.6^2} \approx 37.52$

56. From $E(a) = 0$
$f(a) - g(a) = 0 \rightarrow f(a) = g(a) = c$

From $\lim_{x \to a} \frac{E(x)}{x-a} = 0 \Rightarrow \lim_{x\to a} \frac{f(x)-f(a)}{x-a} - \lim_{x \to a} \frac{g(x)-g(a)}{x-a} = 0$
$f'(a) - g'(a) = 0 \rightarrow f'(a) = g'(a) = m$

$\therefore\ g(x) = m(x-a) + c = f'(a)(x-a) + f(a)$

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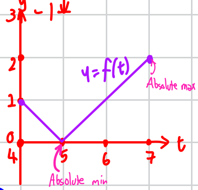
1. The graph has an absolute maximum at $x=b$ and absolute minimum at $x=c_2$, since $h$ is continuous on $[a,b]$, there exists both extreme values by Theorem 1.
2. The graph has an absolute maximum at $x=c$ and absolute minimum at $x=b$, since $f$ is continuous on $[a,b]$, there exists both extreme values by Theorem 1.
3. The graph has an absolute maximum at $x=c$ but does not have absolute minimum since the domain is an open interval, Theorem 1 does not work on it and there is no need to have absolute extreme values.
4. The function is not continuous on $[a,b]$, by Theorem 1, there is no need to have both absolute extreme values. And we can see that this graph does not have any extreme values.
5. The graph has absolute maximum and absolute minimum at $x=a$ and $x=c$. By Theorem 1, there may not be an absolute extreme values, but we get $f(a) \le f(x) \le f(x_c)$ $\forall x \in [a,b]$, so $x=a$ and $x=c$ are the location of absolute extreme values.
6. Since the domain is half-open interval, by theorem 1, it does not guarantee to have absolute extreme values. However, in this graph, the absolute maximum is at $x=a$ and the absolute minimum is at $x=c$.

20.


Absolute maximum: None (Function is half-open on the interval, so it does not satisfy Theorem 1)

Absolute minimum: $(-1,0)$, $(\frac{\pi}{2},0)$

36.




Absolute maximum: $(7,2)$

Absolute minimum: $(5,0)$

42. $f(x) = 6x^2 - x^3$

$f'(x) = 12x - 3x^2$

$f'(x) = 0$ when $12x - 3x^2 = 0$

$3x(4-x) = 0 \rightarrow x = 0, 4$

Critical points are $x=0$ and $x=4$

45. $y = x^2 + \frac{2}{x}$

$y' = 2x - \frac{2}{x^2} = 0$

$\frac{2(x^3-1)}{x^2} = 0 \rightarrow x = 1$

$\frac{2(x^3-1)}{x^2}$ undefined at $x=0$ (which $y$ is also not defined at that point so $x=0$ is not in domain)

$\therefore$ The only critical point is at $x=1$.

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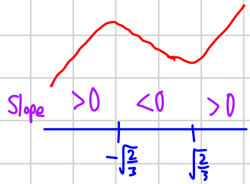
50. $y = x^3 - 2x + 4$

$y' = 3x^2 - 2 = 0 \rightarrow x' = \pm\sqrt{\frac{2}{3}}$




$\therefore$ No absolute maximum/minimum

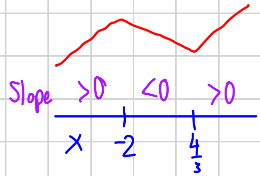
$x = -\sqrt{\frac{2}{3}} \rightarrow$ Local maximum

$x = \sqrt{\frac{2}{3}} \rightarrow$ Local minimum

51. $y = x^3 + x^2 - 8x + 5$

$y' = 3x^2 + 2x - 8 \overset{set}{=} 0$

$(3x-4)(x+2) = 0 \rightarrow x = \frac{4}{3}, -2$




$\therefore$ No absolute maximum/minimum

$x = -2 \rightarrow$ Local maximum

$x = \frac{4}{3} \rightarrow$ Local minimum

67. $f'(x) = \frac{2}{3\cdot\sqrt[3]{x-2}}$

a) $f'(2)$ DNE $\left(\lim_{x\to 0^-} f'(x) = -\infty,\ \lim_{x\to 0^+} f'(x) = \infty\right)$

$f(2)=0,\ f(x)>0,\ x\neq 2$

b) $f'(x) = \frac{2}{3\cdot\sqrt[3]{x-2}}$ is defined and nonzero for $x \neq 2$, hence the only critical point is $x=2$.

c) No, $f(x)$ does not need to have global maximum because its domain is all real numbers. The maximum and minimum values only apply to the interval $[a,b]$

d) $f'(a)$ also does not exist and the $f'(x)$ is defined and nonzero for $x \neq a$, so $x = a$ is its only critical point.

68. $f(x) = \left((x^3 - 9x)^2\right)^{1/2}$

$f'(x) = \frac{1}{2\sqrt{(x^3-9x)^2}} \cdot 2(x^3-9x)(3x^2-9) = (3x^2-9)\cdot\frac{x^3-9x}{|x^3-9x|}$

a) $f(0)$ DNE $\left(\lim_{x\to 0^-} f'(x) = -9,\ \lim_{x\to 0^+} f'(x) = 9\right)$

b) $f'(3)$ DNE $\left(\lim_{x\to 3^-} f(x) = -18,\ \lim_{x\to 3^+} f(x) = 18\right)$

c) $f'(-3)$ DNE $\left(\lim_{x\to -3^-} f(x) = -18,\ \lim_{x\to -3^+} f(x) = 18\right)$

d) $f'(x) = 0$ when $3x^2 - 9 = 0 \rightarrow x = \pm\sqrt{3}$

Hence, all critical points are $x = \underbrace{0, -3, 3}_{\text{Local minimum}}, \underbrace{-\sqrt{3}, \sqrt{3}}_{\text{Local maximum}}$

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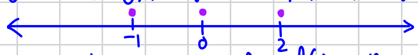
13. The function has a hole at its endpoint $(x=1)$, thus the function is not continuous on $[0,1]$. Rolle's theorem cannot be used on this function.

15. a., iii) $y = (x+1)(x-2)^2 = x^3 - 3x^2 + 4 \rightarrow$ Zeros : $x=-1, x=2$

$y' = 3x^2 - 6x = 0 \rightarrow 3x(x-2) = 0 \rightarrow x = 0, 2$ as critical points




b) Let $x_1, x_2$ be zeros of $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1 x + a_0$ with $x_1 < x_2$

As $f(x_1) = f(x_2) = 0$ and $f(x)$ is continuous on $[x_1, x_2]$, we get that there exists $c \in [x_1, x_2]$ such that $f'(c) = 0$ or $nc^{n-1} + (n-1)a_{n-1}c^{n-2} + \dots + 2a_2 c + a_1$

$\rightarrow$ $c$ is the zero of $nx^{n-1} + (n-1)a_{n-1}x^{n-2} + \dots + a_1$

18. If $f(x)$ is a cubic polynomial with at least 4 zeros, it means $f'(x)$ has at least 3 zeros, $f''(x)$ has at least two zeros and $f'''(x)$ has at least one zero. This contradicts with the fact that $f'''(x)$ must be nonzero constant.

20. As $f(-4) = (-4)^3 + \frac{4}{(-4)^2} + 7 = -56\frac{3}{4} < 0$ and $f(-1) = (-1)^3 + \frac{4}{(-1)^2} + 7 = 10 > 0$, by IVT, we get that there exists $x \in (-4,-1)$ or $x \in (-\infty, 0)$ such that $f(x) = 0$.

Uniqueness of zero : let $c_1, c_2$ be the different zeros of $f(x)$ such that $c_1, c_2 \in (-\infty, 0)$

By Rolle's theorem, we get that there exists $c \in (c_1, c_2)$ such that $f'(c) = 0$

$\therefore f'(c) = 3x^2 - \frac{8}{x^3} = 0$

$\frac{3x^5 - 8}{x^5} = 0 \rightarrow 3x^5 = 8 \rightarrow x = \sqrt[5]{\frac{8}{3}} \notin (-\infty, 0)$

$\therefore$ There exists exactly one zero of $f(x)$ in $(-\infty, 0)$.

23. $r(\theta) = \theta + \sin^2\left(\frac{\theta}{3}\right) - 8$

Since $r(\pi) = \pi + \sin^2\left(\frac{\pi}{3}\right) - 8 = \pi - \frac{29}{4} < 0$ and $r(3\pi) = 3\pi + \sin^2(\pi) - 8 = 3\pi - 8 > 0$, by IVT, it can be seen that there exists $x \in (\pi, 3\pi)$ or $x \in (-\infty, \infty)$ such that $r(x) = 0$.

Uniqueness of zero : Let $c_1, c_2$ be zeros of $r(\theta)$ such that $c_1, c_2 \in (-\infty, \infty)$

By Rolle's theorem, we get that there exists $c \in (c_1, c_2)$ such that $r'(c) = 0$

$r'(c) = 1 + \frac{2}{3}\sin\frac{c}{3}\cos\frac{c}{3} = 0$

$\therefore \sin\frac{c}{3}\cos\frac{c}{3} = -\frac{3}{2}$

Since $|\sin\frac{c}{3}| \leq 1$ and $|\cos\frac{c}{3}| \leq 1$, we get $|\sin\frac{c}{3}\cos\frac{c}{3}| \leq 1 \cdot 1 = 1$ which contradicts with $\sin\frac{c}{3}\cos\frac{c}{3} = -\frac{3}{2}$.

Hence, there exists exactly one zero of $r(\theta)$ in $(-\infty, \infty)$.

27. Since $f'(x) = 0$ for all $x$, we get that $f(x)$ is continuous on every $[x_1, x_2]$ and differentiable at every point. By Mean Value Theorem, there exists $c \in [x_1, x_2]$ such that $\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) = 0 \therefore f(x_2) = f(x_1)$ for all $x_1, x_2$. Hence, $f(x)$ is constant and since $f(-1) = 3$, we get $f(x) = 3$ for all $x$.

33. a) $y' = -\frac{1}{x^2} \rightarrow y = \frac{1}{x} + C$ ; $C$ const

b) $y' = 1 - \frac{1}{x^2} \rightarrow y = x + \frac{1}{x} + C$ ; $C$ const

c) $y' = 5 + \frac{1}{x^2} \rightarrow y = 5x - \frac{1}{x} + C$ ; $C$ const

Section 4.2 (Continue)

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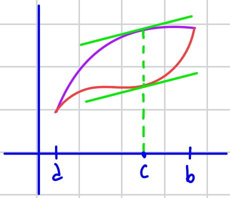
60. Since $f$ and $g$ are differentiable on $[a,b]$, $f$ and $g$ are also continuous on $[a,b]$.
Let $k(x)=f(x)-g(x)$, we get that $k$ is also differentiable and continuous on $[a,b]$.
As $k(a)=f(a)-g(a)=0=f(b)-g(b)=k(b)$, by Rolle's Theorem, there exists $c\in[a,b]$ such that $k'(c)=0 \rightarrow f'(c)-g'(c)=0 \rightarrow f'(c)=g'(c)$
Hence, there exists a point between $a$ and $b$ where the tangents of $f$ and $g$ are parallel or the same line.




61. By MVT, we get that $\exists c\in[1,4]$, $\frac{f(4)-f(1)}{4-1}=f'(c)\leq 1$
$\therefore f(4)-f(1)\leq 1\cdot 3=3$

63. Since $\cos x$ is continuous and differentiable everywhere, by MVT, for all $x$, there exists $c\in[0,x]$ such that $\frac{\cos x-\cos 0}{x}=-\sin c$
$$\frac{|\cos x-\cos 0|}{|x|}=|-\sin c|\leq 1$$
$$\therefore |\cos x-1|\leq |x| \quad \blacksquare$$

65. As $f'(x)-g'(x)=0$ for all points, by corollary 1, it can be implied that $f(x)-g(x)=C$ for some constant $C$. Since $f(x)$ and $g(x)$ starts at the same point, it means $f(x)-g(x)=0$, in other words, $f$ and $g$ are identical.

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Section 4.3

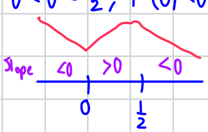
15. Increasing: $(-2,0)\cup(2,4)$, Decreasing: $(-4,-2)\cup(0,2)$
Absolute Maximum: $(-4,2)$ Local Maximum: $(0,1)$, $(4,-1)$
Absolute Minimum: $(2,-3)$ Local Minimum: $(-2,0)$

16. Increasing: $(-4,-\pi)\cup(-\frac{\pi}{2},1)\cup(2,4)$, Decreasing: $(-\pi,-\frac{\pi}{2})\cup(1,2)$
Absolute Maximum: $(4,2)$ Local Maximum: $(-\pi,1)$, $(1,1)$
Absolute Minimum: $(-\frac{\pi}{2},-1)$ Local Minimum: $(2,0)$

17. Increasing: $(-4,-1)\cup(\frac{1}{2},2)\cup(2,4)$ Decreasing: $(-1,\frac{1}{2})$
Absolute Maximum: $(4,3)$ Local Maximum: $(2,1)$, $(-1,2)$
Absolute Minimum: None Local Minimum: $(\frac{1}{2},-1)$, $(-4,-1)$

18. Increasing: $(-4,-\frac{5}{2})\cup(-1,1)\cup(3,4)$ Decreasing: $(-\frac{5}{2},-1)\cup(1,3)$
Absolute Maximum: None Local Maximum: $(-\frac{5}{2},1)$, $(1,2)$, $(4,2)$
Absolute Minimum: None Local Minimum: $(-1,0)$, $(3,1)$

23. $f(\theta)=3\theta^2-4\theta^3 \rightarrow f'(\theta)=6\theta-12\theta^2=6\theta(1-2\theta)$
$\therefore f'(\theta)>0$ (Increasing) when $0<\theta<\frac{1}{2}$, $f'(\theta)<0$ (Decreasing) when $\theta<0$ or $\theta>\frac{1}{2}$
$f'(\theta)=0$ at $\theta=0,\frac{1}{2}$




No absolute maximum and absolute minimum
Local maximum: $(\frac{1}{2},\frac{1}{4})$, Local minimum: $(0,0)$

Section 4.3 (Continue)

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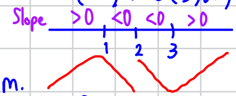
35. $f(x) = \frac{x^2-3}{x-2} = x+2+\frac{1}{x-2} \longrightarrow f'(x) = 1-\frac{1}{(x-2)^2} \quad ; x \neq 2$

$\therefore f'(x) < 0$ (Decreasing) when $|x-2| < 1 \quad \therefore x \in (1,2) \cup (2,3)$

$f'(x) > 0$ (Increasing) when $|x-2| > 1 \quad \therefore x \in (-\infty, 1) \cup (3, \infty)$

$f'(x) = 0$ when $|x-2| = 1 \rightarrow x = 1, 3$

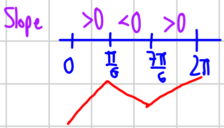



- No absolute maximum and absolute minimum.
- Local maximum : $(1, 2)$ , Local minimum : $(3, 6)$

55. a) $f(x) = \sqrt{3}\cos x + \sin x \quad ; 0 \le x \le 2\pi$

$f'(x) = -\sqrt{3}\sin x + \cos x \overset{\text{set}}{=} 0$

$\cos x = \sqrt{3}\sin x \longrightarrow \tan x = \frac{1}{\sqrt{3}} \quad \therefore x = \frac{\pi}{6}, \frac{7\pi}{6}$



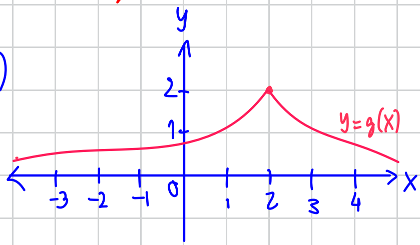

Critical points : $(0, \sqrt{3}) \rightarrow$ Local minimum

$(\frac{\pi}{6}, 2) \rightarrow$ Absolute maximum

$(\frac{7\pi}{6}, -2) \rightarrow$ Absolute minimum

$(2\pi, \sqrt{3}) \rightarrow$ Local maximum

65. a)




b)